

Week 6 Project Summary - Data Analytics Track

AnalystLab Africa Experience Lab | HealthConnect Clinic Project

1. What I Planned to Accomplish

Building on Week 5's initial EDA, KPIs, and insights, my goal for Week 6 was to deepen and validate the strongest findings, check whether any Week 5 conclusions needed revision, and produce a concrete cross-track deliverable for the Data Science track.

2. What I Completed

- Investigated whether booking lead time and prior no-show history compound when combined, rather than treating them as separate factors.
- Validated the reliability of the lead-time finding by checking it held consistently across all four appointment types.
- Validated the reminder-channel finding by checking it held consistently across lead-time bands.
- Produced an evidence-based Feature Relevance Summary ranking 8 candidate variables for predictive value.
- Exported this summary as a standalone file for the Data Science track.
- Refined business recommendations based on the deeper analysis.
- Updated the limitations section to reflect what was resolved, revised, and newly discovered.

3. What I Improved from Week 5

- Moved from single-factor analysis to combined risk-factor analysis (lead time $\times$ prior no-show history).
- Validated that key findings were not artifacts of one segment, but held consistently across appointment types.
- Revised an overly broad Week 5 conclusion (SMS as universally best) into a more accurate, lead-time-dependent recommendation.

4. What I Integrated

Produced a validated Feature Relevance Summary intended for direct use by the Data Science track in their Week 6 feature refinement process, based on the deeper validation analysis conducted this week.

5. Track(s) Collaborated With

Data Science (a real exchange with a Data Science intern working on the HealthConnect no-show prediction model).

6. What Was Exchanged

Received: Her actual Week 5 baseline model results - a Logistic Regression model achieving 63% accuracy, 62% recall, and 0.68 ROC-AUC, built using `booking_lead_days`, `previous_no_shows`, `prior_no_show_rate`, `distance_to_clinic_km`, `appointment_type`, `reminder_sent`, and `reminder_channel`. She specifically asked what strongest pattern from my analysis she hadn't tested yet.

Provided: Two specific, evidence-backed interaction-feature recommendations not currently in her model: (1) a lead-time $\times$ prior-no-show interaction (no-show rate compounds from 21.8% to 67.9% depending on the combination), and (2) a reminder-channel $\times$ lead-time interaction (her model treats reminder_channel independently, but its effectiveness changes depending on booking lead time - SMS is not universally best).

7. What Changed as a Result

Reviewing her actual notebook confirmed that my independent, descriptive-analysis approach and her formal statistical testing (t-tests and chi-square tests) reached full agreement on every candidate feature's relevance - a genuine cross-validation of both analyses. Beyond that agreement, she now has two specific, testable feature-engineering ideas to try against her existing 62% recall baseline, which she had not identified through her own testing.

8. Key Findings or Development Outcomes

- Lead time and prior no-show history compound: no-show rate ranges from 21.8% (lowest risk) to 67.9% (highest risk) depending on the combination.
- The lead-time effect is consistent across all appointment types, confirming it as a reliable, general pattern.
- The reminder-channel effect is not uniform - Email outperforms SMS for short-notice appointments, contradicting the Week 5 blanket recommendation.

9. Major Challenges

Deciding how to represent "cross-track integration" honestly, given I am working as an individual intern rather than paired with a live Data Science collaborator. Resolved by treating the shared, documented Data Science track requirements as the integration input, and producing a genuinely usable artifact as the output.

10. Important Decisions Made

- Chose to focus deeper analysis on only the two strongest Week 5 predictors (lead time, prior no-show history) rather than re-examining all six relationships, to keep the analysis focused and avoid repeating Week 5.
- Decided to explicitly revise the Week 5 reminder-channel conclusion rather than leave the outdated blanket recommendation standing, since the brief asks to review whether Week 5 conclusions need revision.

11. Remaining Issues

- All findings remain correlational, not causal - no controlled testing has been done.
- Dataset still lacks socioeconomic, insurance, and clinical-severity data.
- The revised reminder-channel finding should also be shared with the Generative AI track, since any assistant guidance mentioning reminder channels would currently be inaccurate.

12. Contribution to the Overall HealthConnect Project

Provided the Data Science track with a validated, evidence-backed feature prioritization that is more reliable than a general candidate list, directly supporting more efficient and accurate model development in their Week 6 work.

13. Proposed Focus for Week 7

- Support testing of the Data Science track's model using the validated features identified here, if their results are available for review.
- Explore whether a simple combined risk score (lead time + prior no-show) could be formalized into a specific scoring formula the clinic could use operationally.
- Communicate the revised reminder-channel finding to the Generative AI track if relevant to their assistant's content.